

Implementing High-Security REMOTE CONTROL SYSTEMS



Cristian Toma is senior applications engineer, MCU08 Division, Microchip Technology Inc.

In remote-control operated systems, security is always a concern. The system must be designed to prevent unauthorized access. Special software and encryption are used to prevent eavesdropping. This is why such systems are using encrypted communication. In a perfect scenario, only the authorized receiving party is able to decode and process the information. With an adequate level of encryption, such criteria are generally met. Still, there is the possibility of copying and re-transmitting a valid data transmission. Today, different methods are used to prevent this. However, none of these are perfect.

This article describes the implementation of a highly-secure remote control system that makes the capture and reply attack useless. The data packet sent over the air contains time-synchronization information, which is valid only for a very narrow time frame. Any further capture and reply attack is simply useless. Even though a captured-and-replied message, in essence, contains (almost) correct information, the message is discarded since it has lost its valid time frame. The following paragraphs describe the proposed solution and the technical challenges faced while designing such a system.

The end application rules. General remote-control designs come in various implementations and security levels. Of course, in general, the price also increases along with the security level. However, with smart firmware and by taking advantage of the ever-increasing number of peripherals a modern microcontroller has onboard, designing a remote-controlled system with very strong security is more cost-effective, today.

In general, the remote-control level of security is tied directly with the end

application. For example, controlling a light source or a fan does not really require encryption. But, when the system is designed to allow (or not) access to a valuable property, then there is the need to add a certain security level. However, this does not come for free.

Authentication versus privacy

There is a wide range of encryption algorithms that can be used. The choice is more system-specific. In general, there is a split between authentication and security. Authentication is used more when the encrypted data does not carry any specific information by itself such as commands and messages. It is used more to authenticate a device. A good example is a manufacturer choosing to authenticate replacement parts, upon installation. The system first authenticates these, and only if found valid does it start normal operation. Typically, such devices include printer cartridges, toothbrush heads, mobile phone batteries or even electronic modules.

Algorithms to secure your message.

These devices normally use hash algorithms such as Secure Hash Algorithm-2 (SHA-2) cryptographic hash functions. These functions calculate a key from a longer message. For example, MD5 checksum is used when downloading large files from the Internet. MD5 message digest algorithm will read the entire file and then generate a key. This computed hash value is later used to verify the integrity (or authenticity) of the downloaded file, without providing any means to recover the original data. In other words, simply having the key does not mean you can backwards generate the original data from which the key was generated. In general, in these kinds of applications, authentication is for securing business rather than providing privacy.